

One-Step Equations with Rational Coefficients

one-step equations: undo adding or subtracting, divide both sides, multiply by the reciprocal, or undo a division

Directions:

- Think of every equation as a *balance*: whatever you do to one side, do to the other, so the pans stay level.
- Solve each equation for x , showing the undo step you chose, and check your solution in the original equation.

1. Solve. *Balance model*: the left pan holds x and 7 unit weights, the right pan holds 12 unit weights — remove 7 from *both* pans.

$$x + 7 = 12$$

Answer: _____

2. Solve. *Tape model*: the whole strip is x , cut into 5 equal parts, and each part is worth 4.

4	4	4	4	4
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$$\frac{x}{5} = 4$$

Answer: _____

3. Solve: $x - \frac{3}{4} = 2$

Answer: _____

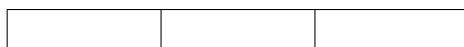
4. Solve, writing the answer as a fraction: $6x = 21$

Answer: _____

5. Solve: $0.4x = 10$

Answer: _____

6. Solve. *Bar model:* the bar x is cut into 3 equal parts, and 2 of those parts together total 8.



$$\frac{2}{3}x = 8$$

Answer: _____

7. Solve: $\frac{x}{2.5} = 6$

Answer: _____

8. Solve, writing the answer as a fraction in simplest form: $\frac{3}{4}x = \frac{9}{10}$

Answer: _____

9. Solve: $x + 2.6 = 1.4$

Answer: _____

10. **Challenge.** Two-fifths of a number x is 18.

(a) Write an equation that says this in symbols.

Answer: _____

(b) Solve your equation to find the number.

Answer: _____